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MILITARY SPECIFICATION

PACKING MATERIAL, METALLIC, FLEXIBLE (SYMBOL 1430)

All interested Bureaus of the Navy Department have concurred in the use of this specification.

1. SCOPE

1.1 This specification covers flexible metallic packing (Symbol 1430) for use on reciprocating rods, rotary rods, valve stems, and outside packed plungers.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
- NN-B-591 - Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
- NN-B-601 - Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
- NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
- QQ-S-781 - Strapping, Flat; Steel.
- ZZ-R-601 - Rubber Goods; General Specifications (Methods of Physical Tests and Chemical Analyses).
- LLL-B-631 - Boxes, Fiber, Corrugated (for Domestic Shipment).
- LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment).

MILITARY
- JAN-P-103 - Packaging and Packing for Overseas Shipment - Boxes, Wood-Cleated, Solid-Fiberboard.
- JAN-P-105 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated Plywood.
- JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
- JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

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MILITARY (cont'd.)
JAN-P-120 - Packaging and Packing for Overseas Shipment - Cartons, Folding, Paperboard.
JAN-B-121 - Barrier-Material, Greaseproof.
JAN-P-133 - Packaging and Packing for Overseas Shipment - Boxes, Set-Up, Paperboard.
JAN-P-139 - Packaging and Packing for Overseas Shipment - Plywood Container Grade.
MIL-L-10547 - Liners, Case, Waterproof.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

POST OFFICE DEPARTMENT
Postal Regulations.

(Application for copies should be addressed to the Post Office Department, Washington 25, D.C.)

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Qualification. - Packing material furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4 (see 6.2).

3.2 Construction and composition. - The packing material shall be square or slightly rectangular in cross section. It shall be made of soft pliable metal foil, ribbon, or strands, wrapped, crimped, twisted, or braided into a continuous strip, with or without a central core of nonmetallic material, in such a manner that the entire working surface will be metal. When a core is used, it shall be concentric in cross section. The construction and composition of the packing material shall conform to that of the sample submitted for qualification.

3.3 Size. - The packing material shall be furnished in the sizes specified (see 6.1).

3.4 Metal. -

3.4.1 Heat resistance. - The metal shall not melt or disintegrate at or below 550°F. (287.78°C.) when tested as specified in 4.5.1.

3.4.2 The metal shall be soft and pliable. It shall not chip or flake off by reason of the abrasive action of the rod when the packing material is tested as specified 4.5.4.

3.5 Lubricant. - The finished packing material shall contain not more than 10 percent of lubricating material and graphite (see 4.5.2).

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3.6 Flexibility.- The packing material shall be sufficiently flexible to be readily formed into coils to fit a rod, the diameter of which is three times the thickness of the packing material (see 4.5.3).

3.7 Simulative service.- The packing material shall pass the simulative service tests specified in 4.5.4. On conclusion of the tests the packing material shall be in satisfactory condition, shall be pliable, and shall have a smooth metallic wearing surface. The rods shall not be grooved, cut, or scored.

3.8 Workmanship.- The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Qualification test at a Government laboratory.- Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.5.

4.3 Sampling.-

4.3.1 Inspection lot.- For purposes of sampling, a lot shall consist of all finished packing material of one size, produced under essentially the same conditions, and offered for delivery at one time.

4.3.2 Sampling for lot acceptance inspection.- A random sample of lengths of packing material shall be selected from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on single sampling inspection requirements in accordance with table I and the procedures of Standard MIL-STD-105.

Table I - Sampling for visual and dimensional inspection
AQL (approx.) = 2.5 percent defective.

Number of lengths of packing material in inspection lot	Number of lengths of packing material in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	2	3
501 to 800	50	3	4
801 to 1,300	75	4	5
1,301 and over	110	6	7

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4.3.3 Sampling for lot acceptance tests at place of manufacture. - A random sample of lengths of packing material shall be selected from each inspection lot in accordance with table II. From each sample length the Government inspector shall take a 24-inch specimen for the tests specified in 4.4.2.

Table II - Sampling for lot acceptance tests.

Number of lengths of packing material in inspection lot	Number of lengths of packing material in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	5	0	1
16 to 40	7	0	1
41 to 110	10	0	1
111 to 300	15	0	1
301 to 500	25	1	2
501 to 800	35	1	2
801 to 1,300	50	2	3
1,301 and over	75	3	4

4.3.4 Sampling for production check test at a Government laboratory. - From the first lot of packing material of each size offered for delivery under a contract or order, the Government inspector shall select two lengths of packing material at random. From each sample length, a 24-inch specimen shall be cut off for the purpose of the production check test specified in 4.4.3. Thenceforth, two such sample pieces of packing material shall be selected from one of every ten lots which have passed inspection and tests at the place of manufacture. Unless otherwise specified in the contract or order, tests shall be conducted at the U.S. Naval Engineering Experiment Station, Annapolis, Md.

4.4 Inspection and tests. -

4.4.1 Visual and dimensional inspection. - Each of the sample specimens selected in accordance with 4.3.2 shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Minor surface defects not affecting the serviceability of the packing material shall not be cause for rejection. Any length of packing material in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective lengths of packing material in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming lengths of packing material. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

4.4.2 Lot acceptance tests at place of manufacture. - The sample specimens selected in accordance with 4.3.3 shall be subjected to the tests specified in 4.5.1 and 4.5.3. Any length of packing material in the sample that fails one or more tests shall be rejected, and if the number of defective sample lengths exceeds the acceptance number, the lot represented by the sample shall be rejected.

4.4.2.1 Action in case of failure. - A rejected lot may be resubmitted for Government inspection only after the manufacturer, after being informed of the reasons for rejection, has so reworked the entire lot as to remove or correct all nonconforming material.

4.4.3 Production check tests at a Government laboratory. - The sample specimens selected in accordance with 4.3.4 shall be subjected to those tests which will establish the identity of the inspection samples with that given qualification and to the tests specified in 4.5.1 to 4.5.3, inclusive.

4.4.3.1 Action in case of failure. - Acceptance of the first lot offered for delivery under a contract or order shall be withheld until a satisfactory report is received on the production check test sample. Thenceforth, except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the lot acceptance inspection and tests specified in 4.4.1 and 4.4.2. Lot acceptance shall then not be withheld pending receipt of a test report on the subsequent production samples. However, upon

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receipt of an unsatisfactory test report on a production check test sample, the Government inspector shall select additional samples from every subsequent lot offered for delivery. The sample so selected shall be submitted to the U.S. Naval Engineering Experiment Station, and shall there be subjected to the production check tests specified in 4.4.3. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. Additional testing shall be discontinued and lot acceptance returned to the normal basis when four successive lots have been accepted.

4.5 Test procedures.-

4.5.1 Heat resistance of metal.- A specimen shall be cut from each of the samples and subjected to a dry heat temperature of 550°F. for 2 hours.

4.5.2 Lubricant.- The lubricant shall be determined as specified in Specification ZZ-R-601. A 5-gram specimen should be used.

4.5.3 Flexibility.- The packing shall be coiled by hand around a rod, the diameter of which shall be three times the thickness of the packing. Only sufficient packing to make two complete turns around the rod shall be used.

4.5.4 Simulative service test.- The simulative service test shall be made under simulated operating conditions in a machine designed for the purpose. The following test conditions shall apply:

Reciprocating rod service with saturated steam at 300 pounds per square inch and a rod speed of 100 feet per minute. The test shall be concluded after 3500 hours of operation.
Rotary rod service with feed water at 100 p.s.i. gage and 250°F., and a peripheral rod speed of 4000 feet per minute. The test shall be concluded after 1500 hours of operation.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment, immediate known use.- Preservation and packaging shall be in accordance with manufacturer's commercial practice.

5.1.2 For domestic shipment and storage or overseas shipment.- Lengths of metallic packing, as specified in the contract or order, shall be wrapped in greaseproof barrier material conforming to type II, grade A of Specification JAN-B-121. The metallic packing shall then be placed in snug-fitting unit and intermediate containers complying with Specification JAN-P-120, JAN-P-133, LLL-B-631 or LLL-B-636. The gross weight of folding cartons or set-up boxes shall not exceed 10 pounds, and of fiberboard boxes shall not exceed 45 pounds. Closure and sealing of the boxes shall conform to the applicable container specification.

5.2 Packing.-

5.2.1 For domestic shipment, immediate known use.- The metallic packing, packaged as specified in 5.1.1, shall be packed in accordance with the latest edition of the Consolidated Freight Classification Rules or Postal Regulations, whichever may be applicable. When fiberboard is used for container construction, the Mullen test shall be no less than 200 pounds. The gross weight of wood boxes shall not exceed 200 pounds.

5.2.2 For domestic shipment and storage.- Unless otherwise specified in the contract or order, the metallic packing, packaged as specified in 5.1.2, shall be packed one or more intermediate containers in snug-fitting wood cleated fiberboard, cleated plywood, nailed wood, corrugated or solid fiberboard boxes conforming to Specification NN-B-591, NN-B-601, NN-B-621, LLL-B-631, or LLL-B-636, respectively. Fiberboard boxes shall conform to the special requirements of the applicable box specification. Closure of the shipping container shall conform to the applicable container specification. The gross weight of wood boxes shall not exceed 150 pounds, and of fiberboard boxes shall not exceed 70 pounds.

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5.2.3 For overseas shipment.- The metallic packing, packaged as specified in 5.1.2, shall be packed by placing one or more intermediate containers in snug-fitting wood cleated fiberboard, wood cleated plywood, nailed wood, or fiberboard boxes conforming to Specification JAN-P-103, JAN-P-105, style 2, 2-1/2, or 3 of Specification JAN-P-106, or symbol V3c or V3s of Specification JAN-P-108, respectively. Plywood shall conform to type A or B, condition I of Specification JAN-P-139. Boxes shall be lined with a sealed waterproof case liner conforming to type I, grade B, class 2 of Specification MIL-L-10547. Seams and closures shall conform to the appendix of Specification MIL-L-10547. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. Flat steel strapping shall conform to class A or B of Specification QQ-S-781. The gross weight of wood boxes shall not exceed 150 pounds, and of fiberboard boxes shall not exceed 70 pounds.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages and intermediate and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129, and with the date of manufacture. The nomenclature shall be: Packing Material, Metallic, Flexible (Symbol 1430).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 3.3).
(c) Whether packing for domestic or overseas shipment is required, if domestic, the type required (see 5.1 and 5.2).

6.2 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of Ships, Navy Department, Washington 25, D.C., and arrange to have the products that they propose to offer to the Army, the Navy, or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Navy Department, Washington 25, D.C.

6.3 The packaging, packing, and marking specified herein apply only to direct shipment to the Government and are not intended to apply to contracts or orders between the manufacturer and the prime contractor.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Other interest:
Or

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